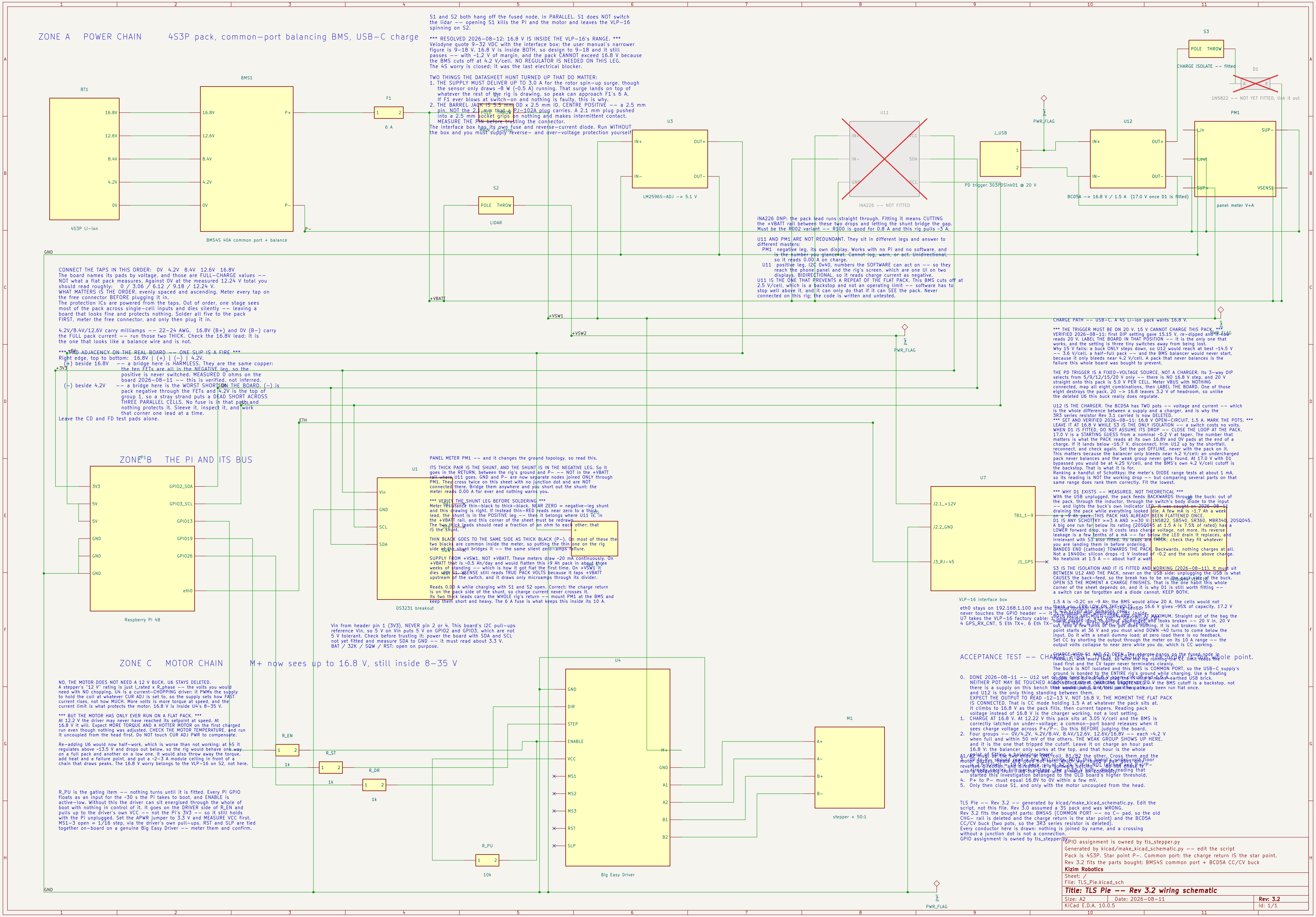

ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

*** RESOLVED 2026-08-12: 16.8 V IS INSIDE THE VLP-16's RANGE. ***
Velodyne quote 9-32 VDC with the interface box; the user manual's narrower figure is 9-18 V. 16.8 V is inside BOTH, so design to 9-18 and it still passes -- with ~1.2 V of margin, and the pack CANNOT exceed 16.8 V because the BMS cuts off at 4.2 V/cell. NO REGULATOR IS NEEDED ON THIS LEG.
The 4S worry is closed; it was the last electrical blocker.

TWO THINGS THE DATASHEET HUNT TURNED UP THAT DO MATTER:

1. THE SUPPLY MUST DELIVER UP TO 3.0 A for the rotor spin-up surge, though the sensor only draws ~8 W (~0.5 A) running. That surge lands on top of whatever the rest of the rig is drawing, so peak can approach F1's 6 A. If F1 ever blows at switch-on and nothing is faulty, this is why.
2. THE BARREL JACK IS 5.5 mm OD x 2.5 mm ID, CENTRE POSITIVE -- a 2.5 mm pin, NOT the 2.1 mm that a PJ-102A plug carries. A 2.1 mm plug pushed into a 2.5 mm socket grips on nothing and makes intermittent contact. MEASURE THE PIN before trusting the connector.

The interface box has its own fuse and reverse-current diode. Run WITHOUT the box and you must supply reverse- and over-voltage protection yourself.

CONNECT THE TAPS IN THIS ORDER: 0V 4.2V 8.4V 12.6V 16.8V
The board names its pads by voltage, and those are FULL-CHARGE values -- NOT what a flat pack measures. Against 0V at the measured 12.24 V total you should read roughly: 0 / 3.06 / 6.12 / 9.18 / 12.24 V.
WHAT MATTERS IS THE ORDER, evenly spaced and ascending. Meter every tap on the free connector BEFORE plugging it in.
The protection ICs are powered from the taps. Out of order, one stage sees most of the pack across single-cell inputs and dies silently -- leaving a board that looks fine and protects nothing. Solder all five to the pack FIRST, meter the free connector, and only then plug it in.

4.2V/8.4V/12.6V carry milliamps -- 22-24 AWG. 16.8V (B+) and 0V (B-) carry the FULL pack current -- run those two THICK. Check the 16.8V lead: it is the one that looks like a balance wire and is not.

*** PAD ADJACENCY ON THE REAL BOARD -- ONE SLIP IS A FIRE ***
Right edge, top to bottom: 16.8V | (+) | (-) | 4.2V.
(+) beside 16.8V -- a bridge here is HARMLESS. They are the same copper: the ten FETs are all in the NEGATIVE leg, so the positive is never switched. MEASURED 0 ohms on the board 2026-08-11 -- this is verified, not inferred.
(-) beside 4.2V -- a bridge here is the WORST SHORT ON THE BOARD. (-) is pack negative through the FETs and 4.2V is the top of group 1, so a stray strand puts a DEAD SHORT ACROSS THREE PARALLEL CELLS. No fuse is in that path and nothing protects it. Sleeve it, inspect it, and work that corner one lead at a time.
Leave the CD and FD test pads alone.

INA226 DNP: the pack lead runs straight through. Fitting it means CUTTING the +VBATT rail between these two drops and letting the shunt bridge the gap. Must be the R002 variant -- R100 is good for 0.8 A and this rig pulls ~3 A.

U11 AND PM1 ARE NOT REDUNDANT. They sit in different legs and answer to different masters:
PM1 negative leg, its own display. Works with no Pi and no software, and is the number you glance at. Cannot log, warn, or act. Unidirectional, so it reads 0.00 A on charge.
U11 positive leg, I2C 0x40, numbers the SOFTWARE can act on -- so they reach the phone panel and the rig's screen, which are one UI on two displays. BIDIRECTIONAL, so it reads charge current as negative.
U11 IS THE ONE THAT PREVENTS A REPEAT OF THE FLAT PACK. This BMS cuts off at 2.5 V/cell, which is a backstop and not an operating limit -- software has to stop well above it, and it can only do that if it can SEE the pack. Never connected on this rig; the code is written and untested.

CHARGE PATH -- USB-C. A 4S Li-ion pack wants 16.8 V.

*** THE TRIGGER MUST BE ON 20 V. 15 V CANNOT CHARGE THIS PACK. ***
VERIFIED 2026-08-11: first DIP setting gave 15.15 V, re-dipped and it now reads 20 V. LABEL THE BOARD IN THAT POSITION -- it is the only one that works, and the setting is three tiny switches away from being lost.
Why 15 V fails: a buck ONLY steps down, so U12 would reach at best ~14.5 V -- 3.6 V/cell, a half-full pack -- and the BMS balancer would never start, because it only bleeds near 4.2 V/cell. A pack that never balances is the failure this whole board was bought to prevent.

THE PD TRIGGER IS A FIXED-VOLTAGE SOURCE, NOT A CHARGER. Its 3-way DIP selects from 5/9/12/15/20 V only -- there is NO 16.8 V step, and 20 V straight onto this pack is 5.0 V PER CELL. Meter VBUS with NOTHING connected, map all eight combinations, then LABEL THE BOARD. One of those eight destroys the pack. 20 -> 16.8 leaves 3.2 V of headroom, so unlike the deleted U6 this buck really does regulate.

U12 IS THE CHARGER. The BCD5A has TWO pots -- voltage and current -- which is the whole difference between a supply and a charger, and is why the 3R3 series resistor Rev 3.1 carried is now DELETED.
*** SET AND VERIFIED 2026-08-11: 16.8 V OPEN-CIRCUIT, 1.5 A. MARK THE POTS. ***
LEAVE IT AT 16.8 V WHILE S3 IS THE ONLY ISOLATION -- a switch costs no volts.
WHEN D1 IS FITTED, DO NOT ASSUME ITS DROP -- CLOSE THE LOOP AT THE PACK.
17.0 V is a STARTING GUESS from a nominal ~0.2 V at taper. The number that matters is what the PACK reads at its own 16.8V and 0V pads at the end of a charge. If it lands below ~16.7 V, disconnect, trim U12 up by the shortfall, reconnect, and check again. Set the pot OFFLINE, never with the pack on it.
This matters because the balancer only bleeds near 4.2 V/cell: an undercharged pack never balances and the weak group never gets found. At 17.0 V with D1 bypassed you would be at 4.25 V/cell, and the BMS's own 4.2 V/cell cutoff is the backstop. That is what it is for.
Ranking a handful of Schottkys: the meter's DIODE range tests at about 1 mA, so its reading is NOT the working drop -- but comparing several parts on that same range does rank them correctly. Fit the lowest.

*** WHY D1 EXISTS -- MEASURED, NOT THEORETICAL ***
With the USB unplugged, the pack feeds BACKWARDS through the buck: out of the pack, through the inductor, through the switch's body diode to the input -- and lights the buck's own indicator LED. It was caught on 2026-08-11 draining the pack while everything looked idle. A few mA is ~1.7 Ah a week on a ~9 Ah pack. THIS PACK HAS ALREADY BEEN FLATTENED ONCE.
D1 IS ANY SCHOTTKY >=3 A AND >=30 V: 1N5822, SB540, SR360, MBR340, 20SQ045.
A big one run far below its rating (20SQ045 at 1.5 A is 7.5% of rated) has a LOWER forward drop, so it costs less charge voltage, not more. Its reverse leakage is a few tenths of a mA -- far below the LED drain it replaces, and irrelevant with S3 also fitted. Its leads are THICK; check they fit whatever you are landing them in before ordering.
BANDED END (cathode) TOWARDS THE PACK. Backwards, nothing charges at all.
Not a 1N400x: silicon drops ~1 V instead of ~0.2 and the sums above change.
No heatsink at 1.5 A -- about half a watt.

S3 IS THE ISOLATION AND IT IS FITTED AND WORKING (2026-08-11). It must sit BETWEEN U12 AND THE PACK, never on the USB side: unplugging the USB is what CAUSES the back-feed, so the break has to be on the pack side of the buck.
OPEN S3 THE MOMENT A CHARGE FINISHES. That is the one habit this whole corner of the sheet depends on, and it is why D1 is still worth fitting -- a switch can be forgotten and a diode cannot. KEEP BOTH.

1.5 A is ~0.2C on ~9 Ah; the BMS would allow 20 A, the cells would not thank you. ERR LOW ON THE VOLTS -- 16.6 V gives ~95% of capacity, 17.2 V is 4.3 V/cell and damages cells.
BOTH POTS ARE MULTI-TURN AND SHIP AT MAXIMUM. Straight out of the bag the board passes input to output unchanged and looks broken -- 20 V in, 20 V out, and a few turns of the pot does nothing. It is not broken: the set point starts at 36 V and you must wind DOWN ~40 turns to come below the input. Do it with a small dummy load; at zero load there is no feedback.
Set CC by shorting the output through the meter on its 10 A range -- the output volts collapse to near zero while you do, which is CC working.

CHARGE WITH S1 AND S2 OPEN. The charger hangs on the fused node in PARALLEL with every load, so with the rig running the CC limit feeds the load first and the CV taper never terminates cleanly.
The buck is NOT isolated and this BMS is COMMON PORT, so the USB-C supply's ground is bonded to the ENTIRE rig's ground while charging. Use a floating supply, and do not also plug the Pi into a mains-earthed USB brick.
DO NOT LEAVE IT CHARGING UNATTENDED -- the BMS cutoff is a backstop, not the control loop, and this pack has already been run flat once.

PANEL METER PM1 -- and it changes the ground topology, so read this.

ITS THICK PAIR IS THE SHUNT, AND THE SHUNT IS IN THE NEGATIVE LEG. So it goes in the RETURN, between the rig's ground and P- -- NOT in the +VBATT rail where U11 goes. GND and P- are now separate nodes joined ONLY through PM1. They cross twice on this sheet with no junction dot and are NOT connected there. Bridge them anywhere and you short out the shunt: the meter reads 0.00 A for ever and nothing warns you.

*** VERIFY THE SHUNT LEG BEFORE SOLDERING ***
Meter resistance thin-black to thick-black. NEAR ZERO = negative-leg shunt and this drawing is right. If instead thin-RED reads near zero to a thick lead, the shunt is in the POSITIVE leg -- then it belongs where U11 is, in the +VBATT rail, and this corner of the sheet must be redrawn.
The two thick leads should read a fraction of an ohm to each other: that IS the shunt.

THIN BLACK GOES TO THE SAME SIDE AS THICK BLACK (P-). On most of these the two blacks are common inside the meter, so putting the thin one on the rig side of the shunt bridges it -- the same silent zero-amps failure.

SUPPLY FROM +VSW1, NOT +VBATT. These meters draw ~20 mA continuously. On +VBATT that is ~0.5 Ah/day and would flatten this ~9 Ah pack in about three weeks of standing -- which is how it got flat the first time. On +VSW1 it dies with S1. VSENSE still reads TRUE PACK VOLTS because it taps +VBATT upstream of the switch, and it draws only microamps through its divider.

Reads 0.00 A while charging with S1 and S2 open. Correct: the charge return is on the pack side of the shunt, so charge current never crosses it.
Its two thick leads carry the WHOLE rig's return -- mount PM1 at the BMS and keep them short and heavy. The 6 A fuse is what keeps this inside its 10 A.

ZONE B THE PI AND ITS BUS

Vin from header pin 1 (3V3), NEVER pin 2 or 4. This board's I2C pull-ups reference Vin, so 5 V on Vin puts 5 V on GPIO2 and GPIO3, which are not 5 V tolerant. Check before trusting it: power the board with SDA and SCL not yet fitted and measure SDA to GND -- it must read about 3.3 V.
BAT / 32K / SQW / RST: open on purpose.

eth0 stays on 192.168.1.100 and the phone hotspot must not. The sensor never touches the GPIO header -- it is Ethernet the whole way. TB1 inside U7 takes the VLP-16 factory cable: 1 Gnd/Shield, 2 +12 Vdc, 3 GPS_Pulse_CNT, 4 GPS_RX_CNT, 5 Eth TX+, 6 Eth TX-, 7 Eth RX+, 8 Eth RX-, 9 GPS return.

ZONE C MOTOR CHAIN M+ now sees up to 16.8 V, still inside 8-35 V

NO, THE MOTOR DOES NOT NEED A 12 V BUCK. U6 STAYS DELETED.
A stepper's "12 V" rating is just I_rated x R_phase -- the volts you would need with NO chopping. U4 is a current-CHOPPING driver: it PWMs the supply to hold the coil at whatever CUR ADJ is set to, so the supply sets how FAST current rises, not how MUCH. More volts is more torque at speed, and the current limit is what protects the motor. 16.8 V is inside U4's 8-35 V.

*** BUT THE MOTOR HAS ONLY EVER RUN ON A FLAT PACK. ***
At 12.2 V the driver may never have reached its setpoint at speed. At 16.8 V it will. Expect MORE TORQUE AND A HOTTER MOTOR on the first charged run even though nothing was adjusted. CHECK THE MOTOR TEMPERATURE, and run it uncoupled from the head first. Do NOT touch CUR ADJ PWR to compensate.

Re-adding U6 would now half-work, which is worse than not working: at 4S it regulates above ~13.5 V and drops out below, so the rig would behave one way on a full pack and another on a low one. It would also throw away the torque, add heat and a failure point, and put a ~2-3 A module ceiling in front of a chain that draws peaks. The 16.8 V worry belongs to the VLP-16 on S2, not here.

R_PU is the gating item -- nothing turns until it is fitted. Every Pi GPIO floats as an input for the ~30 s the Pi takes to boot, and ENABLE is active-low. Without this the driver can sit energised through the whole of boot with nothing in control of it. It goes on the DRIVER side of R_EN and pulls up to the driver's own VCC -- not the Pi's 3V3 -- so it still holds with the Pi unplugged. Set the APWR jumper to 3.3 V and MEASURE VCC first.
MS1-3 open = 1/16 step, via the driver's own pull-ups. RST and SLP are tied together on-board on a genuine Big Easy Driver -- meter them and confirm.

A1/A2 must be the two ends of ONE coil, B1/B2 the other. Cross them and the motor buzzes, heats and does not turn. Which way round a pair goes only reverses direction, and direction is a software setting -- do not chase it with a soldering iron. Find the pairs with a meter on continuity.

ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

0. DONE 2026-08-11 -- U12 set on the bench to 16.8 V open-circuit and 1.5 A. NEITHER POT MAY BE TOUCHED AGAIN: with the trigger at 20 V there is a supply on this bench that would put 5.0 V/cell on the pack, and U12 is the only thing standing between them.
EXPECT THE OUTPUT TO READ ~12-13 V, NOT 16.8 V, THE MOMENT THE FLAT PACK IS CONNECTED. That is CC mode holding 1.5 A at whatever the pack sits at. It climbs to 16.8 V as the pack fills, then current tapers. Reading pack voltage instead of 16.8 V is the charger working, not a lost setting.
1. CHARGE AT 16.8 V. At 12.22 V this pack sits at 3.05 V/cell and the BMS is correctly latched on under-voltage; a common-port board releases when it sees charge voltage across P+/P-. Do this BEFORE judging the board.
2. Four groups -- 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V -- each ~4.2 V when full and within 50 mV of the others. THE WEAK GROUP SHOWS UP HERE, and it is the one that tripped the cutoff. Leave it on charge an hour past 16.8 V: the balancer only works at the top, and that hour is the whole point of fitting a balancing board.
3. 0V to P- should read a few MILLIvolts. NOTE this board's under-volt floor is 2.5 V/cell = 10.0 V pack, so at 12.24 V it is NOT latched and P+/P- already carries full pack voltage. The ~0.55 V body-diode reading that started this investigation belonged to the OLD board's higher threshold.
4. P+ to P- must equal 16.8V to 0V within a few mV.
5. Only then close S1, and only with the motor uncoupled from the head.

TLS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 3S pack and was WRONG.
Rev 3.2 fits the bought parts: BMS4S (COMMON PORT -- no C- pad, so the old CHG- rail is deleted and the charge return is the star point) and the BCD5A CC/CV buck (two pots, so the 3R3 series resistor is deleted).
Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
GPIO assignment is owned by tls_stepper.py.

GPIO assignment is owned by tls_stepper.py
Generated by kicad/make_kicad_schematic.py -- edit the script
Pack is 4S3P. Star point P-. Common port: the charge return IS the star point.
Rev 3.2 fits the parts bought: BMS4S common port + BCD5A CC/CV buck
Kizim Robotics
Sheet: /
File: TLS_Pie.kicad_sch
Title: TLS Pie -- Rev 3.2 wiring schematic
Size: A2 | Date: 2026-08-11 | Rev: 3.2
KiCad E.D.A. 10.0.5 | Id: 1/1